



QUALITY SYSTEM

FORM

DOCUMENT #: 3.100.019.F01

LEVEL: 3

Effective Date:

APPROVED

JEsqueda , 9/30/2025, 4:15:53 PM

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TITLE:

LABORATORY OOS INVESTIGATION FORM

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OOS Number: OOS- 261187

Phase I

Section A – Test Details

Initiator:	Clea S. Garza (Written by: Qiyue Chen)	Test Date:	11-May-2026	Date Initiated:	18-May-2026
Date of Incident:	18-May-2026	Test Name:	Environmental Monitoring	Sample ID:	ETX-260518-0273
Name of Analyst who Performed the Test:	Clea S. Garza (Changeover Surface Sampling Analyst)	Sample / Active Name	ScanC/O CGS E001309 S1 11MAY2026	Lot #:	ScanC/O CGS E001309 S1 11MAY2026
	Maraya Chukwumerije and Sophia Santamaria (Changeover Surface Sampling Plate Reader)	Dosage Form	Plate		
Description of Incident:	The CFU count for the environmental monitoring plate exceeded the action level.		☐ Client Care notified of OOS ☐ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS		
	SOP / Test Method # MICRO-SOP-2	Effective Date: 23-Jul-2026	SOP / Test Method Rev: 16	Limits/Specification: Action level: >= 1CFU/Plate	
Manager notified	Manager/designee name: Kathan Parikh	Date: 18-May-2026	Quality Assurance Department notified:	Yes ☐ No ☐	

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy		Comments
Analyst interviewed?	☒ Yes ☐ No ☐ N/A	Yes, analysts Clea S. Garza and Maraya Chukwumerije and Sophia Santamaria were comprehensively interviewed.
Correct samples analyzed?	☐ Yes ☐ No ☒ N/A	Not applicable
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-2
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-2
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A	Yes, Information is available on Eagletrax under ETX-260518-0273
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A	Yes, the analysts are trained and qualified by quality to perform the test
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A	Not Applicable
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A	Not Applicable
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A	Yes, as per MICRO-SOP-2
Material Discrepancy		Lot Number
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A	
Was the sample transported appropriately?	☒ Yes ☐ No ☐ N/A	
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A	



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Contact Plate: 1011543730	Contact Plate: 29SEP2026
Standard Used	Not Applicable	Not Applicable	Not Applicable
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A	Please see below	Please see below
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A	CR117 (E001739)	Dec 2026
System Performance Check (if applicable)?	☐ Yes ☐ No ☒ N/A	N/A	N/A
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001034 (Sensor E001501) Incubator E001031 (Sensor E001505)	Aug 2026 Feb 2027 Aug 2026 Feb 2027



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Are chromatogram/spectra/raw data correct?

☐ Yes☐ No☒ N/A

Not Applicable

Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)

☒ N/A☐ Re-injection of sample vial☐ Re-injection of original standard vial☐ Re-vialing of original working standard solution☐ Re-vialing of original working sample solution☐ Re-dilution of a working sample solution from the original stock☐ Re-dilution of a working standard solution from the original stock standard☐ Other: N/A QYC 18-Sep-2026

Section D – Phase I Summary

Phase I Summary:

The analyst involved in the changeover surface sampling plate setup, Clea S. Garza (CGS), and the analysts involved in reading the plate, Maraya Chukwumerije and Sophia Santamaria, were interviewed comprehensively. Their responses are documented throughout this investigation.

The EM plates were stored in accordance with the supplier's recommendations, visually inspected before use, and verified to be within their assigned expiration dates. All materials and supplies were disinfected in accordance with MICRO-SOP-9, Cleaning and Disinfecting Procedure for Microbiology. The functionality of the incubators was verified through review of data generated by the in-house continuous monitoring system.

Changeover Surface Sampling was performed by Clea S. Garza in ISO 5 BSC E001309 located in room 117A in suite CR117 on the date of testing (11 May 2026), in accordance with MICRO-SOP-2, Environmental Monitoring of the Cleanroom Facility.

The plates were initially incubated at a temperature of 30-35°C in incubator E001031 for a minimum duration of 48 hours, commencing on 11 May 2026. Following completion of the minimum 48 hours of incubation on 13 May 2026, no microbial growth was observed. The plates were subsequently incubated for a minimum of 5 days at 20-25°C in incubator E001034, with incubation ending on 18 May 2026. At completion of the second incubation period, one colony-like artifact was observed on S1. The plate was read by Maraya Chukwumerije after the initial incubation period and by Sophia Santamaria after the second incubation period. Please see Table 1 for detailed information on the observations during the respective incubations.

The observed artifact was submitted for microbial identification under ETX-260518-0273. However, following transfer or inoculation onto fresh media, no growth was observed. Therefore, the observed artifact could not be confirmed as a viable microbial colony or reported as a confirmed colony forming unit (CFU). The lack of growth upon subculture indicates that the observation may have been nonviable material, an artifact associated with agar preparation or pouring, or another non-microbial artifact.

To evaluate environmental controls and determine whether the recovery was transient or recurring, personnel-monitoring plates for Clea S. Garza and ISO 5 BSC E001309 environmental-monitoring plates were bracketed across the testing period, as detailed in Table 2.

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Phase I Summary (continued):

Upon analyzing the environmental monitoring results summarized in Table 2, personnel monitoring plates for Clea S. Garza, including left- and right-touch plates, showed no microbial growth on the date before testing (08 May 2026), the date of testing (11 May 2026), and the date after testing (12 May 2026).

For ISO 5 BSC E001309, daily surface sampling of four locations showed no microbial growth on the date before testing (09 May 2026). On the date of testing (11 May 2026), 1 CFU was recovered from S1 associated with Clea S. Garza. The recovery was documented under ETX-260518-0273; microbial identification indicated a colony-like artifact (White, shiny, hardened, smooth area embeded on agar). Surface sampling performed on the date after testing (12 May 2026) showed no microbial growth across all locations.

Settling sampling of ISO 5 BSC E001309, including two locations, showed no microbial growth on the date before testing (09 May 2026), the date of testing (11 May 2026), and the date after testing (12 May 2026).

Weekly active-air monitoring of Suite 117 conducted during the week before testing (29 April 2026) showed no microbial growth. Weekly active-air monitoring conducted during the week of testing (07 May 2026) recovered 2 CFU in Room 117 under ETX-260518-0263, which were identified as Gram-positive cocci and Gram-positive pleomorphic rods. Weekly surface monitoring of the anteroom and cleanroom areas associated with Suite 117 showed no microbial growth on 29 April 2026 or 07 May 2026.

During the interview, the analyst indicated that no obvious abnormalities or deviations occurred during the testing process. All materials were disinfected before testing, and the relevant cleanroom and ISO 5 BSC were cleaned and prepared before testing in accordance with MICRO-SOP-2 and MICRO-SOP-9.

Monthly cleaning and disinfection of the cleanroom suite, including the ISO 8 anteroom (Room 117) and ISO 7 buffer room (Room 117A) housing ISO 5 BSC E001309, were performed on 26-Apr-2026 by Analyst - Tamiru Kotisso and Cuong Du - in accordance with MICRO-SOP-9, Cleaning and Disinfecting Procedure for Microbiology. All H2O2 indicators passed, confirming the successful completion and effectiveness of the monthly cleaning and disinfection activities within Suite 117. Additionally, routine cleaning and disinfection were performed before and after the testing activity in accordance with MICRO-SOP-9.

It is also important to note that no product samples processed by Clea S. Garza in ISO 5 BSC E001309 on the date of testing (11 May 2026) failed SCAN RDI testing.

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Phase I Summary (continued):

Based on the available evidence, one colony-like artifact was observed on ScanC/O CGS E001309 S1 11MAY2026 in ISO 5 BSC E001309 on the date of testing (11 May 2026). However, the observation could not be confirmed as a viable microbial CFU because no growth was obtained following inoculation onto fresh media. Therefore, the observation represents a nonviable or non-microbial artifact, including a potential artifact associated with agar preparation or the agar-pouring process.

Even if the observed artifact had represented a viable CFU, it would appear to be an isolated event. This assessment is supported by the absence of microbial recovery from Clea S. Garza's personnel-monitoring plates on the date before testing (08 May 2026), the date of testing (11 May 2026), and the date after testing (12 May 2026); the absence of growth from BSC E001309 surface samples collected on the date before testing (09 May 2026) and the date after testing (12 May 2026); and the absence of growth from ISO 5 settling plates throughout the bracketing period.

Although active-air monitoring recovered 2 CFU (Gram-positive cocci and Gram-positive pleomorphic rods) in the surrounding area of Room 117 during the week of testing (07 May 2026), this recovery occurred outside the ISO 5 critical environment of BSC E001309. Furthermore, the ISO 5 Surface S1 observation yielded no growth upon subculture and was confirmed as a non-viable artifact. Therefore, a microbiological relationship between the ISO 5 surface artifact and the cleanroom active-air recovery cannot be established.

The negative ISO 5 settling, personnel, and bracketing surface monitoring results demonstrate that the BSC E001309 critical environment remained in a state of control. The available data do not support migration, persistence, or recurrence of contamination within ISO 5 BSC E001309. Collectively, the evidence supports that the recovery was an isolated, transient, and non-recurring event, while established cleaning, disinfection, and aseptic controls remained effective.

Accordingly, no systemic environmental control deficiencies were identified, and no additional corrective or preventive actions are warranted at this time beyond continued routine environmental monitoring and adherence to approved cleaning, disinfection, and aseptic procedures.



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Has root cause been identified as a possible laboratory error?*Note: Root cause must be conclusive, verifiable, and well-documented.*☒ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☐ No, the OOS result is valid.**Reviewer comment(s):****Prepared by (Print):**

Qiyue Chen

Signature/Date:

Lab Manager/Designee (Print):

Robin Seymour

Signature/Date:

Section E – Quality Disposition**Can investigation be closed?**☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.**Approved by QA Representative (Print):**

Signature/Date:

Table 1: Read Dates & Incubation Observation

ETX Submission ID	Set-up Analyst & Date	Sampling Site & Location	Plate Reading Analyst (≥ 48H)	CFUs Observed after 48 Hour Incubation at 30–35°C (E001031)	Plate Reading Analyst (NLT 5 days)	CFUs Observed after NLT 5-day Incubation at 20–25°C (E001034)	Microbial Identification
ETX-260518-0273	CGS 11MAY 2026	Surface sampling plate	MC	0	SAS	1 CFU on surface #1 sampling plate	N/A

Table 2: Environmental Monitoring Plates for Analyst and Cleanroom Bracketing

Environmental Monitoring (EM) Sampling Site	Frequency	Date (DDMM YYYY)	Analyst (Initials)	Day /Week(s)	Observation*	Environmental Monitoring Plate ETX ID	Microbial ID	Notes
Personnel EM Bracketing								
Personal (Left Touch and Right Touch)	Daily	08MAY 2026	CGS	Date Before Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	11MAY 2026	CGS	Date of Testing	No growth	N/A	N/A	None
Personal (Left Touch and Right Touch)	Daily	12MAY 2026	CGS	Date After Testing	No growth	N/A	N/A	None
Biological Safety Cabinet EM Bracketing Biological Safety Cabinet (BSC) 1309								
Surface Sampling of ISO 5 (4 locations)	Daily	09MAY 2026	VV	Date Before Testing	No growth			None
Surface Sampling of ISO 5 (4 locations)	Daily	11MAY 2026	CGS, DT, GA, VV	Date of Testing	1 CFU on S1 for CGS	ETX-260518-0273	N/A White, shiny, hardened, smooth area embeded on agar	None
Surface Sampling of ISO 5 (4 locations)	Daily	12MAY 2026	PG, DT, HS, ELB, GA, VV	Date After Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	09MAY 2026	VV	Date Before Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	11MAY 2026	CGS, DT, GA, VV	Date of Testing	No growth	N/A	N/A	None
Settling Sampling of ISO 5 (2 locations)	Daily	12MAY 2026	PG, DT, HS, ELB, GA, VV	Date After Testing	No growth	N/A	N/A	None
Weekly Active Air Sampling Bracketing 117								
Active Air Sampling of Cleanrooms	Weekly	29APR 2026	SMO	Week (On of Testing Date)	No growth	N/A	N/A	None
Active Air Sampling of Cleanrooms	Weekly	07MAY 2026	SMO	Week (On or After Testing Date)	2 CFU for 117 Air	ETX-260518-0263	Gram (+) cocci Gram (+) pleomorphic rods	None
Surface Sampling of Anteroom and Cleanroom Bracketing 117								
Surface Sampling of Cleanrooms	Weekly	29APR 2026	SMO	Week (On of Testing Date)	No growth	N/A	N/A	None
Surface Sampling of Cleanrooms	Weekly	07MAY 2026	SMO	Week (On or After Testing Date)	No growth	N/A	N/A	None